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Element enrichment characteristics: Insights from element geochemistry of sphalerite in Daliangzi Pb–Zn deposit, Sichuan, Southwest China

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ABSTRACT

Sphalerite in the Daliangzai Pb-Zn deposit, one of the hundreds of Pb-Zn deposits in South China, was chosen as an case study for geochemical research using in situ laser-ablation inductively coupled mass spectroscopy analysis. Many kinds of minor and trace elements were investigated and subsequently treated by multivariate statistical analysis, especially principal component analysis (PCA). These research works indicate that the enrichments of Cu, Ga, As, Cd, Ge, Ag, Sb, Pb, Ni in sphalerite from different depths of the super-large No. I orebogy mutually imply a lower-temperature hydrothermal process dominating the mineralization in Daliangzi Pb-Zn deposit. A small temperature gradient controls the almost similar element distribution patterns indicated by the sphalerite samples which were enriched in Fe, Mn, Co, Cu, Ge, Ag, and Cd in the lower orebody whereas those in the upper orebody were enriched in Fe, Mn, Co, Cu, Ge, Ag, Cd, plus Ga and In. The enrichment of Fe in sphalerite matrix assists the incorporation of many metals, for example, Mn, Co, Ge, and (Ag + Sb). Some important coupled substitutions are discovered: $4Zn^{2+} \leftrightarrow 2Fe^{2+} + Ge^{4+} + \square$ (where $\square$ denotes a vacancy), $2Zn^{2+} \leftrightarrow Ag^{+}$ $(Cu^{+}) + Sb^{3+}$, and $3Zn^{2+} \leftrightarrow 2Ag^{+}$ $(Cu^{+}) + Ge^{4+}$. Moreover, tetrahedrite–tennantite and jordanite–geocronite solid solution series distinctly feature the sphalerite formed in Daliangzi Pb-Zn deposit which differentiates it from other metallogenic types, especially those related to magmatic activities.

Results of PCA work to the selected sphalerite samples from deposits of skarn, syngenetic massive sulfide, MVT, special Jinding and Daliangzi reveal that significant differences, which are mainly defined by the concentration of Fe, Mn, Co, and depleted In, exist in the composition of sphalerite from different deposit types and lie along a single dimension of PC2*. Such a conclusion bases a framework established by PCA and proposed to study the base-metal deposit types and the relevant metallogenic process that are typical for deposits located in South China.

1. Introduction

Sphalerite is an important host mineral in worldwide Mississippi Valley–type (MVT) lead–zinc deposits. As is widely known, naturally grown minerals do not have uniform chemical composition, and sphalerite is not an exception. As a kind of excellent carrier of many different classes of trace and minor elements, as well as kinds of isotopes, sphalerite samples have often been chosen for chemical experiments to develop further understanding of their formation process and deposits (L'Heureux, 2000; Cook et al., 2009, 2015; Ye et al., 2011; Belissont et al., 2014; Gagnevin et al., 2014; George et al., 2016; Frenzel et al., 2016). The concentration and isotopic composition of trace and minor elements trapped by sphalerite are widely used to characterize the physicochemical conditions during fluid–rock interaction and deposition during the evolution of hydrothermal systems, as well as to identify the origin of the substance (Qian, 1987; Michael et al., 2003; Belissont et al., 2014; Cook et al., 2009, 2015; George et al., 2016; Frenzel et al., 2016, and references therein), all of which are absolutely essential for confirming the metallogenic type of a deposit.

Various kinds of minor and trace elements have been detected in

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sphalerite, including Mn, Fe, Co, Ni, Cu, Ga, Ge, As, Mo, Ag, Cd, In, Sn, Sb, W, Au, Hg, Tl, Bi, and Pb, the existence of which are considetered to be of great formation significance (Vaughan and Craig, 1997; Di Benedetto et al., 2005; Cook et al., 2009). Many dozens of papers have addressed various aspects of sphalerite geochemistry. Some have shown correlations between concentrations of trace and minor elements and temperature, for example, Co and In are concentrated in hypo- and mesothermal ores, whereas Ga and Ge are more common at lower temperature (Möller and Dulski, 1993, 1996; Belissont et al., 2014). Also, numerous papers have contributed to studies of the substitution mechanisms of these elements in the sphalerite lattice, how these elements indicate migration, their reaction with the country rock permeated, the subsequent precipitation of ore-forming fluids, and the relation with sphalerite color, etc. which are indispensable for a metallogenic event (Cook et al., 2009; Pfaff et al., 2011; Ye et al., 2011; George et al., 2016; Frenzel et al., 2016, and references therein). In addition, following the development of modern microanalytical techniques, those points exemplified above are being solved better via these new methods (e.g., in situ laser-ablation inductively coupled mass spectroscopy (LA-ICP-MS), secondary ion mass spectrometry, and thermal ionization mass spectrometry) with more accuracy (i.e. sub-part-per-million-level precision and micrometer-scale spatial resolution) and reliability (Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014; George et al., 2016). Not only do these modern microanalytical techniques provide an opportunity to determine concentrations at detection limits lower than that of the electron microprobe, but they also provide direct and indirect information on whether a given trace element is present within the sulfide matrix or as micro-submicroscopic inclusions inside the mineral crystal.

Dian-Qian-Chuan Triangle region of the South China, as a world-class metallogenic belt, is located on the southwestern margin of the Yangtze Block and contains hundreds of deposits and occurrences. The Daliangzi Pb-Zn deposit, one of the major deposits, having a metallic reserve of 1.8 Mt at an average grade of 11.45% (lead + zinc) and a super-large orebody marked No. I with metallic tonnage > 1 Mt and extending > 300 m from the ground surface to depth (Yuan et al., 2014), is considered to be a typical example which deserves more academic attention. The research work of Ye et al. (2011) showed the systematic chemical variations in sphalerite samples from MVT, magmatic–hydrothermal type, sedimentary exhalative type deposits and the special Jinding deposit located in South China. However, because of the special local geological setting of the Daliangzi deposit, only a few scientific contributions have been published that focus on the variety of compositions in typical minerals within the super-large orebody and its implications to the miraculous formation. In this paper, using the fashionable microanalytical technique (LA-ICP-MS), we aim to constrain the laws of composition variety, reasons for minor and trace element substitution, ranges of solid solutions in natural samples, and the possible geological controls on sphalerite geochemistry in the Daliangzi Pb–Zn deposit. Through detailed discussion, all of these undoubtedly provide us with useful proofs to understand the mineralization process and the ore genesis which is important to build the exploration model.

2. Regional geology

Southern China typically comprises three main terranes (Fig. 1(A))—the Sanjiang fold belt system, the Yangtze Block (YB), and the Cathaysian Terrane (CT)—in terms of the partition of regional tectonics (Ye et al., 2011). Each of them experienced a complicated evolutionary history that is considered to be favorable for formation of different kinds of deposits therein (Zaw et al., 2007). The Daliangzi ore district, located in the Dian-Qian-Chuan Triangle region, is one of the most important metallogenic regions belonging to the YB. It is situated spatially close to the Huize Pb–Zn deposit, both of which are placed near the Xiaojiang fault, which cuts across the continental crust (Fig. 1(B); Liu and Xu, 1996). Its basement is composed of Archeozoic and Mid-Proterozoic crystallographic rocks, with successive sedimentary rocks mainly from the Sinian and Cambrian age forming a cap locally (Zheng et al., 2006; Xu et al., 2007). Especially, the carbonate rock-bearing Sinian strata is the most important wall rocks hosting the most of the base metal deposits in Dian-Qian-Chuan Triangle region (Wang et al., 2002). The SN, NW and NE-treading faults form the tectonic framework in the region, controlling the most magmatic avtivity and occurrence of deposits (Fig. 1B). The Permian Emeishan basalts is the most important magmatic rock developed near the researching unit (Fig. 1B; Zheng and Wang, 1991).

3. Ore geology

The outcropping strata in Daliangzi Pb-Zn include the dolomitic ore-bearing Dengying Formation of the upper Sinian series, sandstone and shale formation of Lower Cambrian, and Quaternary eluvium and alluvium (Fig. 2A). There is an uncomformity at the contact surface between the Sinian and Canbrian rocks. The upper Sinian strata, consisting of limestone and dolomite, host all the orebodies. The faults and fissures formed in the Sinian strata are the ore-controlling factors.

The ore district is located in a graben in which there are many fissures and fractures rather than folding (Zheng and Wang, 1991). The NW-trending faults F_1 and F_{15}, being branches of a regional fault, developed as the boundaries of the graben, control the emplacement of the deposits. Some secondary faults to F_1 and F_{15}, such as the NWW-trending F_6, F_7, F_8, F_{12}, F_{13}, and F_{14}, also play an important role in controlling orebodies. Black fissure zones, created by these faults, belong to tectonic breccia zones that are altered by ore-forming fluids enriched in carbon (including bitumen, plumbago, etc.) and have a close spatial relationship with the mineralization; while the black color caused by carbon along the fractures disappears, the mineralization terminates simultaneously (Yuan et al., 2014). No magmatic rocks crop out in the ore district (Fig. 2).

Daliangzi Pb-Zn deposit consist of two orebodies. Orebody No. I, being of super-large scale, is the largest orebody in the deposit; it is 630 m long, 326 m deep, and 0.81–169.79 m wide; its metallic tonnage is > 1.1 Mt at grades up to 11.45% (Pb + Zn). It was shaped by a big breccia-bearing vein, with a clear boundary to the country rock of dolostone. Orebody No. II, controlled by the F13 fault, lacates at the southeast of No. I orebody (Fig. 2A). The metallic minerals of these orebodies are simple (Fig. 3(A)–(F)), indicated by the most abundant sphalerite followed by galena, pyrite and less chalcopyrite, arsenopyrite, marcasite, freibergite and pyrargyrite. The gangue minerals are dominated by calcite, quartz and dolomite. All bodies contain small amounts of Ge, Ga, Cd, and Ag and these trace elements are thought to be generally trapped by the minerals, i.e., sphalerite and galena (Zheng and Wang, 1991).

Primary ores in all orebodies are massive, disseminated, and veined on the whole. Replacement, crystallization, containment, solid solution separation structures prevail in the ores (Fig. 3(C)–(F)). We conclude that orebodies from the Daliangzi deposit have experienced hydrothermal and oxidized periods. The hydrothermal period, in turn, is composed of pyrite–arsenopyrite–quartz, pyrite–galena–sphalerite–chalcopyrite–quartz–carbonate, and cerussite–heminorphite –limonite–carbonate stages in total according to the mineral sequence (Fig. 3(C)–(F)) identified by field work and microscopic identification. The second stage of the hydrothermal period contributes significantly to the formation of all orebodies.

Wall rock alterations include carbonization, silicification, dolomitization, calcitization, and ferritization, of which both the carbonization and silicification are placed closely to the Pb–Zn mineralization (Yuan et al., 2014).

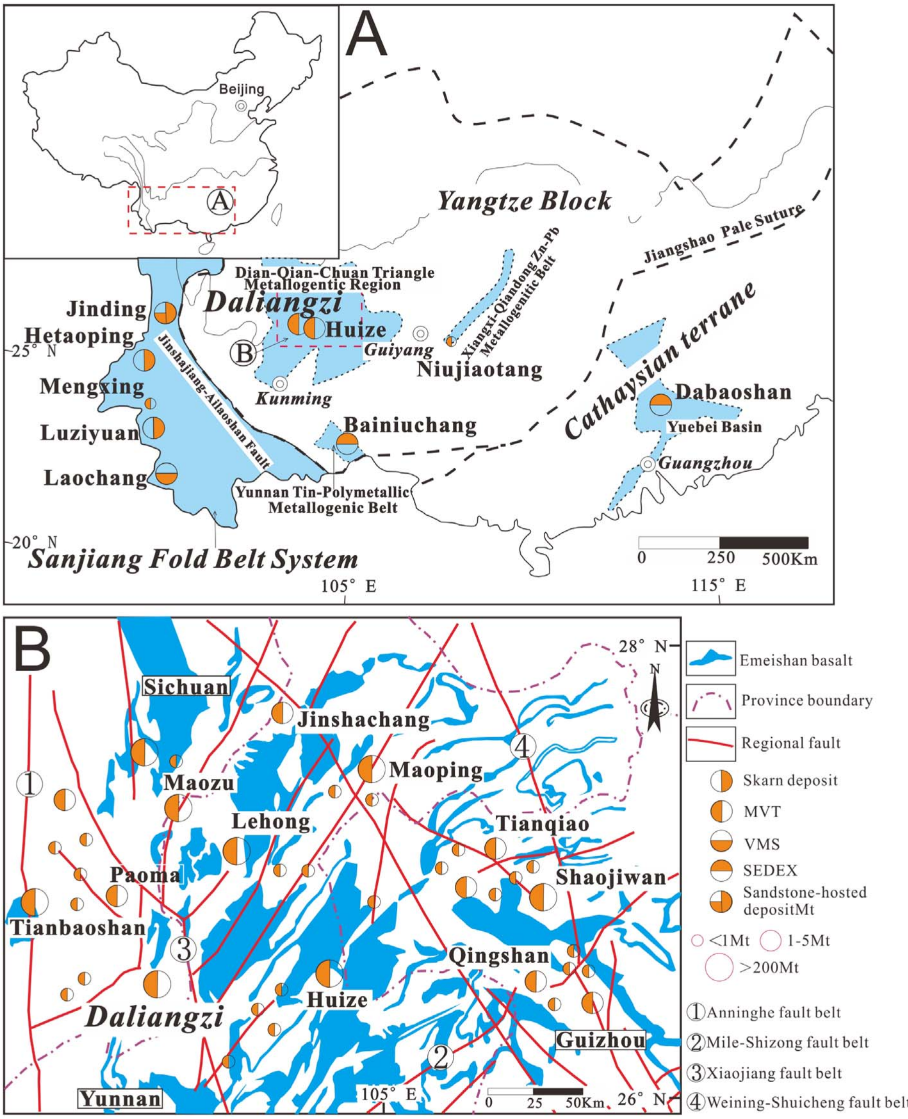

Fig. 1. Geological sketch map (A + B) of South China showing the location and tectonic setting of the deposits studied.
Modified after Ye et al. (2011) and Zhang et al. (2015).

4. Sampling

20 specimens in the main mineralization stage (stage 2 of the hydrothermal period) were collected evenly from different levels of orebody No. I at depths from 1884 to 2064 m (La-1–5 collected from Level 1884-m, La-6–10 from Level 2064-m, La-11–16 from Level 1944-m, and La-17–20 from Level 2004-m, respectively). These 20 specimens, being of similar inside texture, were typical ores dominantly composed of sphalerite, galena, and dolomite. These metallic minerals always occur predominantly as replacement of dolomite or open space fillings (Fig. 3(A) and (B)), with enhedral-subhedral crystal in common. Island-like residue and skeleton crystals of pyrite and galena can be found within sphalerite, which indicates that sphalerite was formed later (Fig. 3(E) and (F)). Generally, these specimens show massive or veinlike structures with typical replacement process within, which indicate that hydrothermal mineralization played a great role in the growth of the orebody and sphalerite is an important production. Additionally, sphalerite is considered to be refractory enough to work as a tracer of metallogenic overprinting (Ye et al., 2011). The sphalerite in these specimen was thus chosen as the research agent to further understand the hydrothermal and mineralized process in Daliangzi Pb-Zn deposit.

In addition, except those from Daliangzi Pb-Zn deposit, datasets of contents of minor and trace elements from sphalerite in the regional deposit of different ore-forming types were collected for comparison with each other (see Electronic Appendix A). Only mean values of the regional deposits were collected, thereby enabling distinct identification and comparison of local or regional trends but ignoring the complex details hiding behind the elusive high-dimensional data.

5. Analytical methods

A series of geochemical tests conducted in the laboratory were arranged for further study, e.g., polished slices were prepared for investigation using an optical electronic microscope, in situ LA-ICP-MS was used to determine the composition of sphalerite, and multivariate statistics presented by principal component analysis (PCA) of in situ datasets was used to discover the relationships among elements. Primary analytical techniques in the laboratory and methods used in this paper are described in the following.

5.1. Laser ablation inductively coupled plasma mass spectrometry

Trace element concentrations were determined by LA-ICP-MS using a 193-nm excimer LA system (ESI NWR UP193-FX) coupled to a quadrupole ICP-MS (PerkinElmer DRC) at the National Research Center for Geoanalysis (China). Ablation was performed using a 40-μm spot size at 23–25 J/m^2. Single spot analyses were ablated using 10 pulses/s. The ablated material was transported via a He carrier gas to a modified

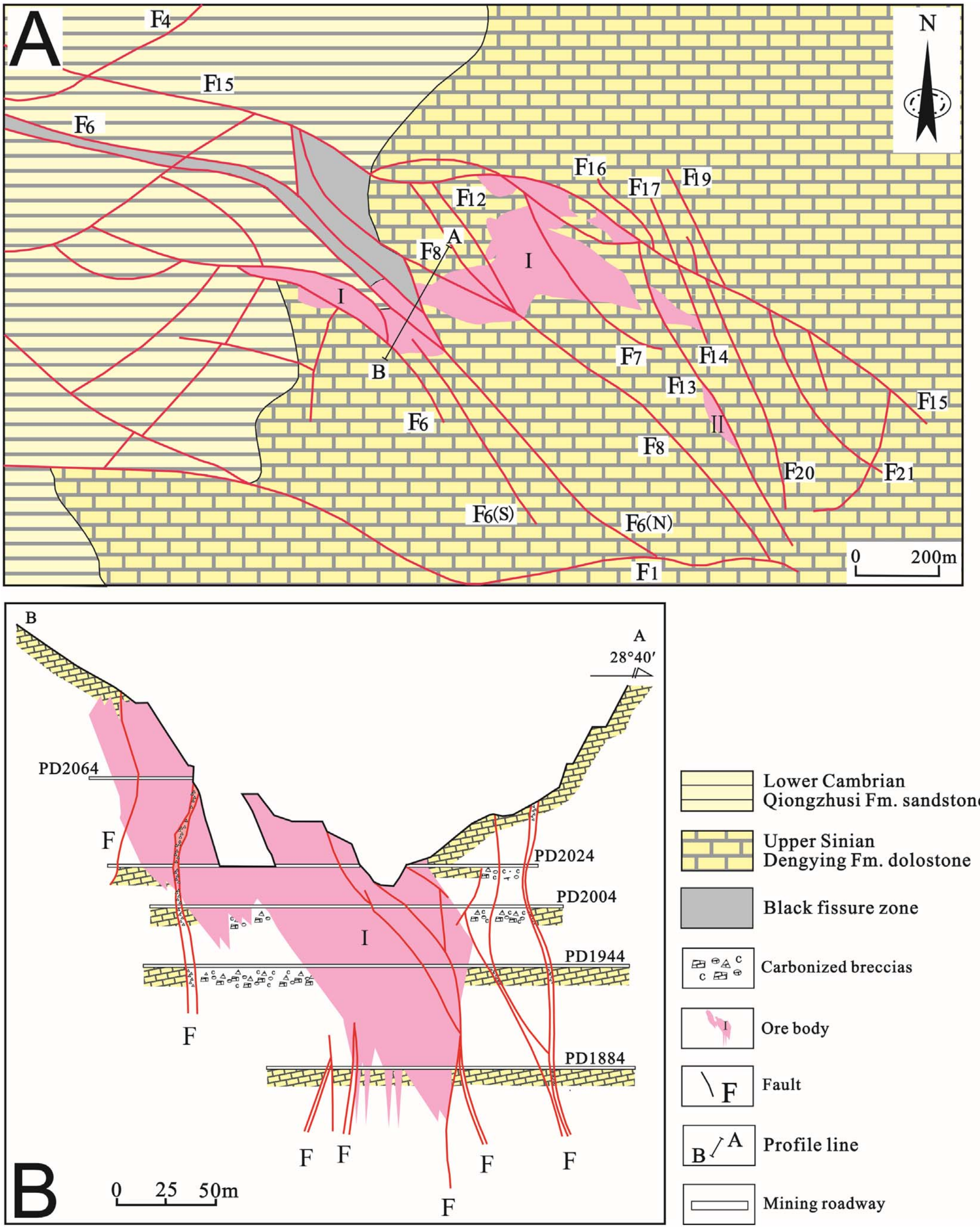

Fig. 2. (A) Geological map of the Daliangzi deposit. (B) Representative cross section A–B through the deposit.
Redrawn after Zhang et al. (2008).

glass mixing bulb where the He^+ sample was mixed coaxially with Ar prior to the ICP torch. The USGS sulfide reference material MASS-1 (Wilson et al., 2002) was used for calibration. The Zn contents as determined by electron microprobe analysis (Zhu et al., 1995; Li, 2003) collected with a standard deviation of 1.3 ppm of 13 spots in all were used as the internal standard. Concentration and detection limit calculations were conducted using the protocol of Longerich et al. (1996). The data are reported in Electronic Appendix A. In addition, great care had to be taken when using the concentration of Ge, because of the poor confidence caused by interference, as shown by Ye et al. (2011). As suggested by Belissont et al. (2014), Ge^{74} is considered to be the least affected isotope, requiring no further corrections; therefore, Ge^{74} was adopted in this study.

5.2. Principal component analysis of in situ datasets

Multivariate statistical analysis is a powerful tool for obtaining a concise description of high-dimensional data highlighting the relationships among variables, especially elements trapped in ores or minerals. One such multivariate statistical analysis, PCA, is commonly used for variable reduction to project data in two dimensions with minimal data loss (Winderbaum et al., 2012; Belissont et al., 2014; Frenzel et al., 2016). Therefore, PCA was thought to be the best method to highlight the most relevant trends without external supervision (e.g., Koch, 2012). Principal components are directly computed as the eigenvectors of the correlation matrix and ordered by decreasing eigenvalues (Belissont et al., 2014; Frenzel et al., 2016). Consequently, the highest variance corresponding to the most significant relationships

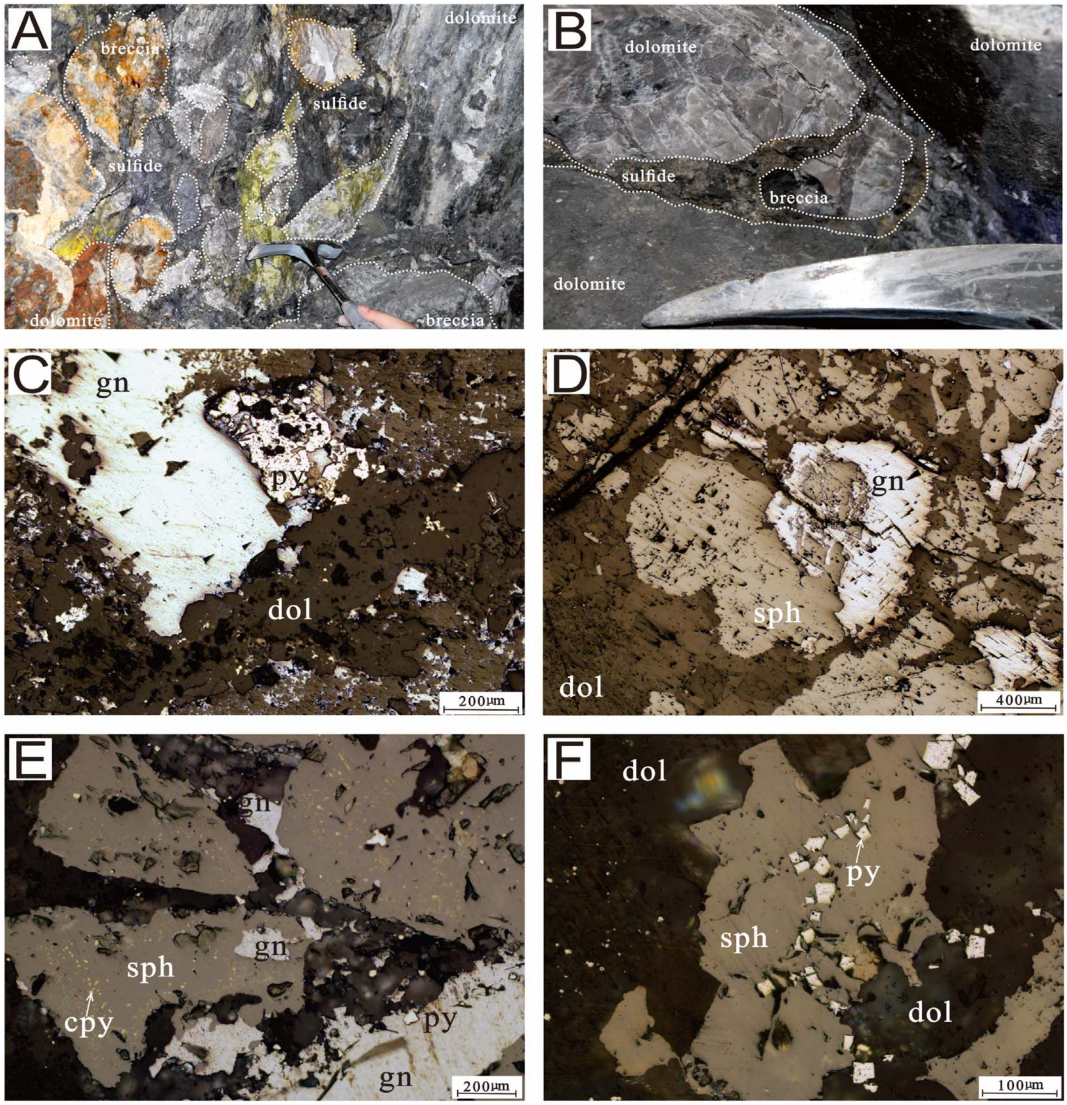

Fig. 3. (A, B) Field photographs of ore in the Daliangzi deposit with vein-type mineralization infilling the void space in breccias and fissures in the country limestone. (C, D, E, F) Representative photomicrographs in reflected light showing ore textures: (C) pyrite replaced by galena with dolomite infilling the space left; (D) galena replaced by sphalerite with dolomite infilling the space between those metallic minerals; (E) galena replaced by sphalerite with intense "chalcopyrite disease" inside; (F) pyrite surrounded by sphalerite. Abbreviations: py = pyrite, cpy = chalcopyrite, sph = sphalerite, dol = dolomite, gn = galena.

between the variables (element concentrations) can be obtained and presented in the way of numbers and histograms.

In this study, PCA is applied to the LA-ICP-MS datasets for its obvious comparative advantage. Not all elements included in the database were selected for such a treatment. Only those satisfying the following criteria were chosen: (1) The element had to have been measured for more than half of all data points; (2) the concentration must have been above the respective detection limit in more than half of all cases; and (3) pioneers could be selected for comparison. In this manner, the minor and trace element contents of Fe, Mn, Cu, Cd, Ga, Ge, In, Co, and Ag were selected through a careful comparison job. Before treating these datasets, first, we considered the processing of values below the detection limit (BDL). Because of the very small proportion ($\ll$10%, mostly) of such values in all, the simplest solution adopted in this paper is imputation with a fixed value to the respective detection limit as common (van den Boogaart and Tolosana-Delgado, 2013). Then, datasets for individual deposit types were compiled into one large dataset and raw LA-ICP-MS concentrations were log-transformed to ensure the approximate normality required for statistical treatment, which was considered to be the standard procedure in the treatment of trace element data (e.g., Gasser, 1974; Winderbaum et al., 2012; Belissont et al., 2014; Frenzel et al., 2016). Then, a PCA module of Origin software (version 9.0) was run on this large dataset and the results were used to assess the higher dimensional nature of the compositional differences in sphalerite samples from different deposit types.

6. Results

6.1. Minor and trace elements

The LA-ICP-MS dataset (85 spot analyses; see Electronic Appendix A for means, standard deviations, and maxima and minima for each sample) shows concentrations of each minor and trace element in natural sphalerite from the Daliangzi deposit. The ranges in absolute concentrations for selected elements are shown as histograms in Fig. 4. Representative time-resolved depth profiles are given in Fig. 5. During the laboratory experiment, we did our best to analyze sphalerite volumes free of obvious inclusions or other features; however, the analyzed samples do nevertheless show extensive inhomogeneity, at least in some samples. This is because the LA-ICP-MS spot size is greater than many of these features, so the ablation profiles will not always show homogeneity on the scale of the ablation spot. Some elements such as

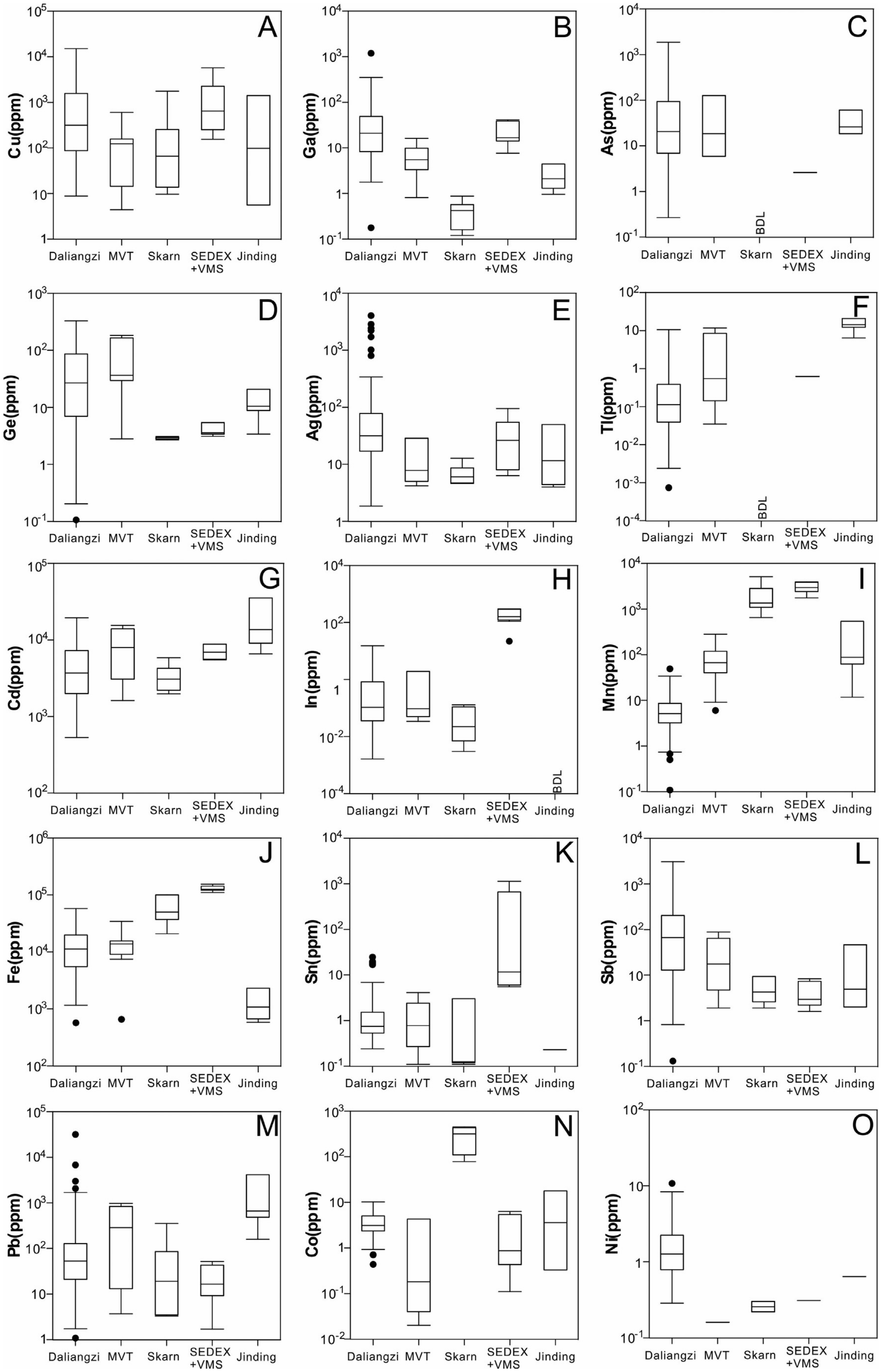

Fig. 4. Trace element contents (LA-ICP-MS) in sphalerite from the Daliangzi deposit and those collected in Ye et al. (2011) varying by MVT, skarn, massive sulfide and special Jinding type. Minimum, maximum, median, lower quartile and upper quartile are indicated by the lowest point, the uppermost point, the line across the rectangular box, the upper boundary of the rectangular box, the lower boundary of the rectangular box of each column, respectively. See text for explanation.

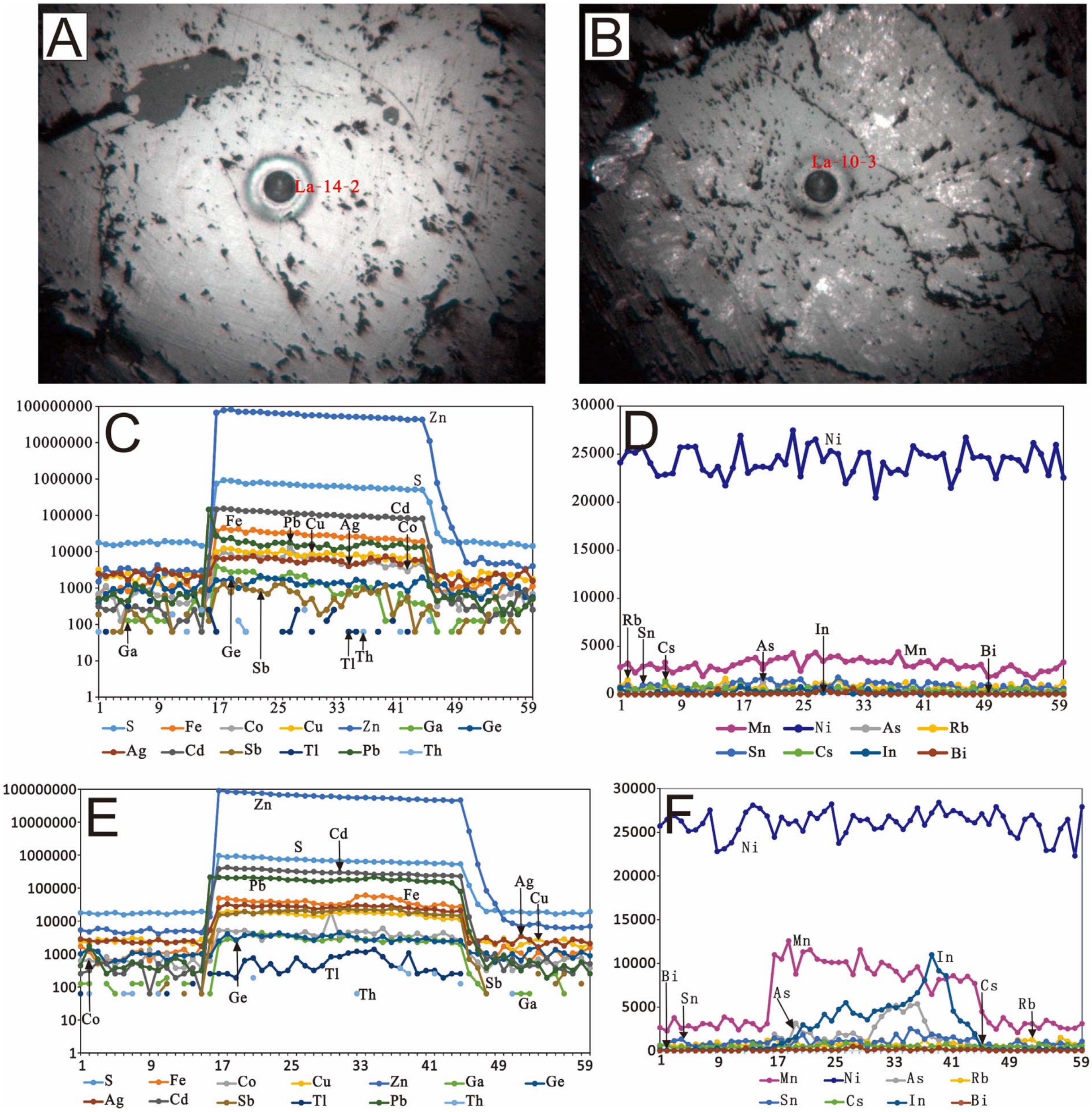

Fig. 5. Representative single-spot (La-14-2 and La-10-3) LA-ICPMS spectra for selected elements in sphalerite. (A, B) Photomicrographs of samples including the spots of La-14-2 and La-10-3 in reflected light. (C, D) LA-ICPMS spectra for the spot of La-14-2. (E, F) LA-ICPMS spectra for the spot of La-10-3.

Fe, Cd, Ag, and Cu always show smooth profiles, suggesting that these are homogenously distributed on the scale of the spot. Other elements, notably Mn, Pb, Ga, Ge, In, Tl, Co, and Sb, commonly—though not always—exhibit irregular profiles, suggesting the presence of micro-scale inclusions of minerals carrying these elements. Such cases can also be indicated if the mean value is smaller than the standard deviation to the concentration of a specific element.

Cu: All specimens contain measurable quantities of copper. A remarkable feature of the dataset for Cu is that concentrations are not constant, varying from 9.1450 to 15,113.3 ppm (with high standard deviations relative to mean value). This feature of Cu concentration can be explained by the type of substitution in sphalerite, including both solid solution and inclusions, some of which are in line with the broadly occurring "chalcopyrite disease." However, some spectra of samples are smooth, and even some inclusions can be readily seen in some mineral crystals during investigation (Fig. 5(C) and (E)). These clearly indicate the presence of solid solutions of Cu in the sphalerite regardless of whether or not inclusions are present. According to Fig. 4(A), the concentrations of Cu in sphalerite from all types are considered at a high level. In the Cu vs. Ge and Cu vs. Sb plots in Fig. 6(A) and (B), the distribution of spots exhibits positive correlations between Cu and Ge and between Cu and Sb.

Fe: Iron concentrations in sphalerite from the Daliangzi deposit range from 0.057 to 5.77 wt%, with a mean value of 1.37 wt% and a standard deviation of 1.12 wt%. Certainly, Fe concentration varies significantly from one sample to another. Compared with the skarn and massive sulfide types, sphalerite samples from the Daliangzi deposit have a lesser volume of Fe (Fig. 4(J)); however, for sphalerite, Fe plays an important role with respect to the texture and also the genetic process. It is surprising that the special Jinding type seems to have the peculiar attribute of having the lowest Fe concentration. The flat and ragged profiles occurring with or without inclusions (Fig. 5(C) and (E)) imply that Fe presents in sphalerite probably via both solid solution and inclusions. Iron concentrations exhibit probable weak positive correlations with Mn, Cd, Ge, Co, and (Ag + Sb) (Fig. 6(C)–(G)).

Ga: The concentrations of gallium in sphalerite from the Daliangzi

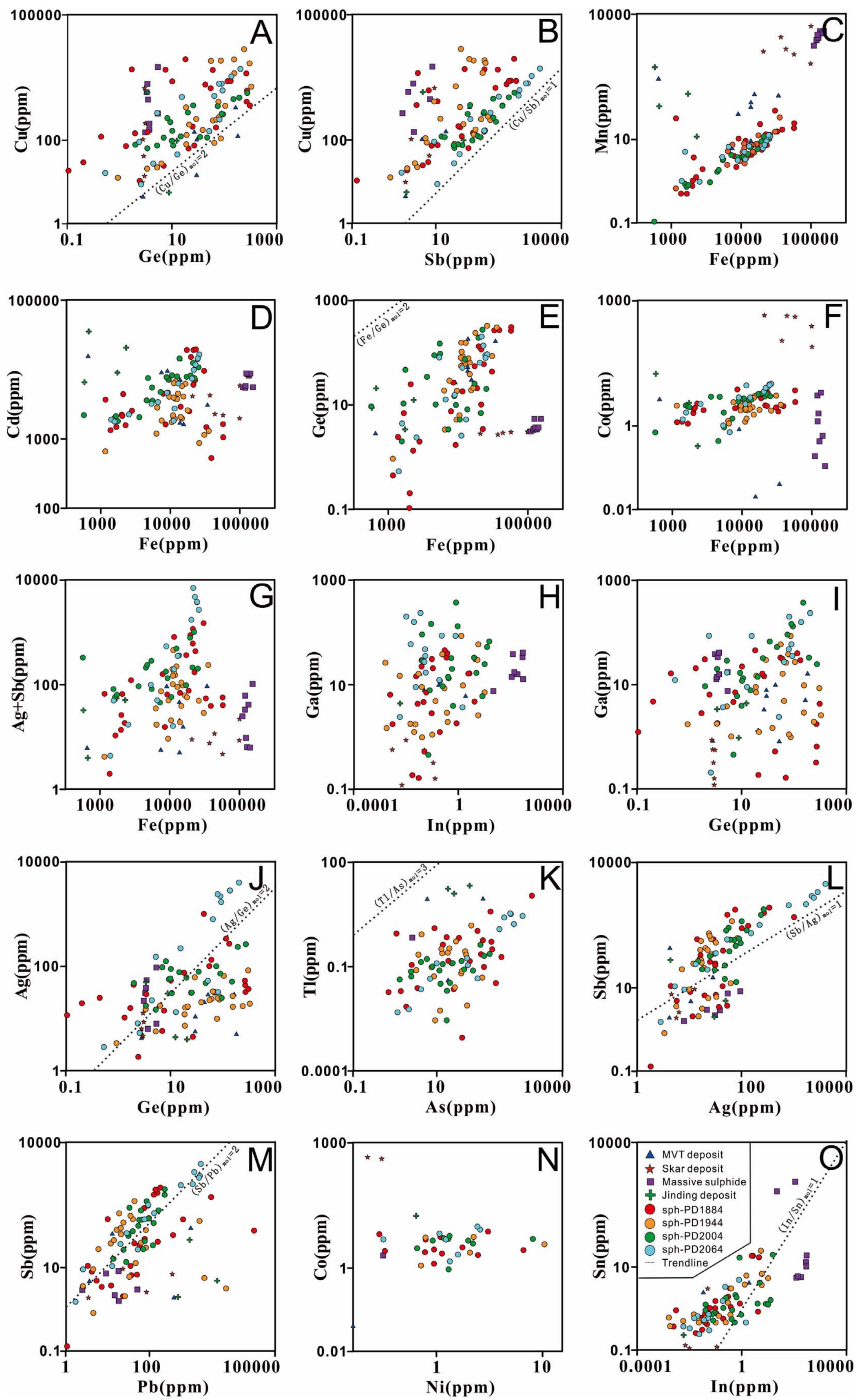

Fig. 6. Correlation plots of (A) Cu vs. Ge, (B) Cu vs. Sb, (C) Mn vs. Fe, (D) Cd vs. Fe, (E) Ge vs. Fe, (F) Co vs. Fe, (G) (Ag + Sb) vs. Fe, (H) Ga vs. In, (I) Ga vs. Ge, (J) Ag vs. Ge, (K) Tl vs As, (L) Sb vs. Ag, (M) Sb vs. Pb, (N) Co vs. Ni and (O) Sn vs. In. Note that the dashed lines represent the theoretical mole ratios implying the potential substitution mechanisms of trace elements trapped in sphalerite crystal. See text for explanation.

deposit are higher than those in other deposits according to Fig. 4(B), with a range of 0.1826–389 ppm (with high standard deviations relative to mean value). In contrast, skarn-type deposits have much lower concentrations of Ga. Although the same genetic type (MVT) can be identified in the Daliangzi, Huize, Mengxing, and Niujiaotang deposits (Zheng and Wang, 1991; Ye et al., 2011), sphalerite from the Daliangzi deposit exhibits a greater potential for containing Ga (Figs. 4(A) and 6(H) and (I)). The spectra of Ga appear jagged commonly; however, smooth ones can also be found though meticulous observation (Fig. 5(C) and (E)).

Ge: It is important to note that we cannot reliably give the absolute values of contents of germanium owing to an analytical problem. Germanium concentrations within sphalerite from the Daliangzi deposit are generally higher than those within other genetic types, according to Fig. 4(D). This feature resembles that of arsenic and gallium. However, concentrations of Ge in sphalerites within the skarn and massive sulfide deposits are significantly lower compared with those of MVT, Jinding type, and Daliangzi deposits. The spectrum of germanium is smooth with very few fluctuations (Fig. 5(C) and (E)). Otherwise, to some extent, Ge concentrations exhibit a positive correlation with Fe, and Ag (Fig. 6(E) and (J)).

As: Time-resolved depth profiles for arsenic are rarely flat for either sphalerite including inclusions or those in which inclusions are absent (Fig. 5(D) and (F)), proving the existence of As-bearing minerals. The trend of this element follows that of gallium, according to Fig. 4(B) and (C), which indicates that sphalerite of the Daliangzi deposit is the most enriched in arsenic. Consistent with the well-known conclusion, As is always higher in comparable, low-temperature replacement-style deposits (Cook et al., 2009; Belissont et al., 2014). It is worth mentioning that the concentrations of arsenic in the skarn-type deposits are notably low (e.g., Hetaoping and Luziyuan), with some being lower than the detection limit (Fig. 4(C)). An unobvious positive correlation between As and Tl can be identified (Fig. 6(K)). Antimony concentrations range widely from tens to thousands of ppm with a mean value of 242.38 ppm and a standard deviation of 472.21 ppm. Antimony concentration in sphalerite from the Daliangzi deposit is the highest compared with other types. The skarn and massive sulfide types correlating with magma processes have the lowest antimony concentrations (Fig. 4(L)). Most antimony spectra are usually ragged with several exceptions (Fig. 5(C) and (E)), with the signal being steady as exemplified by sample La-10–13, in which inclusions are present. Antimony concentrations exhibit a positive correlation with Ag and Pb (Fig. 6(L) and (M)).

Ag: All samples contain measurable quantities of silver. A remarkable feature of the dataset for Ag is the wide range of concentrations (1.84–4037 ppm with a mean value of 225 ppm and a standard deviation of 650 ppm). Compared with those within the MVT and skarn, massive sulfide, and Jinding deposit types, sphalerite from the Daliangzi deposit have the highest concentrations of Ag (Fig. 4(E)), which make it an important byproduct. The spectra of Ag are usually ragged with a few smooth ones in sphalerite either including inclusions or in which inclusions are absent (Fig. 5(C) and (E)), which indicates that Ag must exist mostly as inclusions. In addition, according to the plot of Fig. 6(L), a positive correlation of Sb and Ag can be obtained.

Tl: Thallium concentrations are almost < 1 ppm. This feature of low Tl concentration is similar to those within the skarn and massive sulfide types (Fig. 4(F)). The spectra of thallium are usually ragged in sphalerite with or without inclusions (Fig. 5(C) and (E)), indicating that Tl concentrations are related to microscopic inclusions. In addition, Tl has a weak positive correlation with As (Fig. 6(K)).

Cd: Cadmium is a ubiquitous component within all analyzed sphalerite. The concentrations of Cd is high enough as a byproduct, with a concentration range of 0.055 to 2 wt% (with a mean value of 0.53 wt% and a standard deviation of 0.46 wt%). The spectra of Cd are usually smooth for sphalerite with or without inclusions (Fig. 5(C) and (E)), indicating that concentrations of Cd are related to solid solutions. Some concentrations at the level of 1 wt% probably indicate that the high concentration of Cd may correlate with microscopic inclusions. According to Fig. 4(G), sphalerite from the MVT including the Daliangzi deposit has moderate concentrations of Cd, with abnormally high values occurring in the Jinding deposit and skarn and massive sulfide deposits having relatively lower concentration of Cd. Cadmium has a weak positive correlation with Fe (Fig. 6(D)).

Mn: Manganese concentrations are almost < 10 ppm, indicating that sphalerite from the Daliangzi deposit contains the lowest concentration of Mn compared to the other kinds of deposits (Fig. 4(I)). In reality, the radius of the Mn^{2+} ion is similar to that of Fe^{2+}, and the Mn/Fe ratios in sphalerite from some deposits such as skarn and massive sulfide types are higher than those of MVT deposits including the Daliangzi, indicating that magma processes play an important role in concentrating manganese in sphalerite (Cook et al., 2009; Ye et al., 2011). The Jinding deposit, considered to be a special case, has moderate concentrations of manganese, further differentiating it from the single genetic type such as MVT or sedimentary exhalative (Wang et al., 2009). A stratabound type was widely suggested to describe the genetic type of the Jinding deposit (Wang et al., 2009; Ye et al., 2011). The spectra of Mn are usually ragged, whether or not the sphalerite contains inclusions (Fig. 5(D) and (F)). With a higher standard deviation than a mean value, most of the volume of manganese existing as microscopic inclusions can be identified. Manganese concentration exhibits a positive correlation with Fe (Fig. 6(C)).

Pb: Lead shows a considerable enrichment across the dataset. Pb concentrations lie within the range of 1.08–31,653.3 ppm with a mean value of 679.97 ppm and standard deviation of 3490.81 ppm. The high standard deviation among individual ablation spots and ragged profiles (see Electronic Appendix A; Fig. 5(C) and (E)) imply that Pb is not generally present in solid solution but rather as microscopic inclusions of other sulfides, which is corroborated by the obvious positive correlations with other elements, such as Sb and Ag (Fig. 6(M)), indicating that jordantie–geocronite is probably present as independent microscopic minerals.

Co: Cobalt appears to be systematically enriched in skarn sphalerite relative to other deposit types including the Daliangzi deposit (Fig. 4(N)). Cobalt concentration is low, falling within a narrow range from 0.44 to 10.26 ppm with a mean value of 3.79 ppm and a standard deviation of 2.23 ppm. This concentration feature is similar to those in MVT, massive sulfide, and Jinding type deposits. Most Co spectra are usually ragged, whether or not the sphalerite contains inclusions (Fig. 5(C) and (E)), indicating that most of the Co volume exists as microscopic inclusions. Cobalt concentrations exhibit a positive correlation with Fe and Ni (Fig. 6(F) and (N)).

Ni: Nickel concentrations are low, falling within a narrow range from 0.28 to 10.76 ppm with a mean value of 1.99 ppm and a standard deviation of 2.23 ppm. However, they are the highest among those in the skarn, MVT, massive sulfide, and Jinding type deposits (Fig. 4(O)). The ragged profiles are always present, whether or not the sphalerite contains inclusions (Fig. 5(D) and (F)), proving that nickel is trapped by sphalerite via microscopic inclusions. Nickel concentrations exhibit a weak correlation with Co (Fig. 6(N)).

In: Indium concentrations are summarized as histograms in Fig. 4(H), which shows the features of concentrations for different deposit types. The massive sulfide contains the highest volume of In; in contrast, the concentration of In in sphalerite from the Jinding deposit is the lowest, being almost below the detection limit. The MVT including the Daliangzi deposit has moderate concentration compared to the above two end-members. The spectra of indium are sometimes ragged (Fig. 5(D) and (F)), indicating that some indium exists as microscopic inclusions. This conclusion can be also proved by the fact that the standard deviation is higher than the mean value (see Electronic Appendix A). In addition, there are weak positive correlations between In and Ga and between In and Sn (Fig. 6(H) and (O)).

Sn: Tin concentration in sphalerite from the Daliangzi deposit is

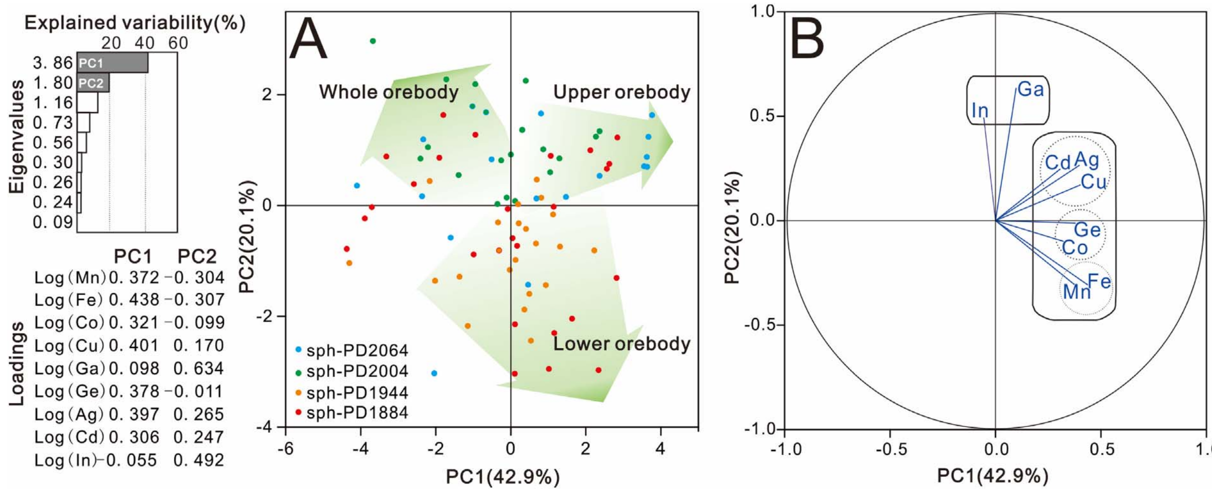

Fig. 7. Principal component analysis of the LA-ICP-MS log-transformed dataset of trace element contents in sphalerite from Daliangzi deposit. On the left hand side edge: eigenvalues (i.e., variance explanation) and loadings (i.e., eigenvector coefficients) of the principal components. Middle frame (A): spot analyses (i.e., individuals) plotted on the PC1 vs. PC2 plane (explaining 63% of element content variability). Right frame (B): elements (i.e., variables) plotted in the same plane. This PCA underlines two main, opposed clusters of elements distributed within sphalerite samples: Fe, Mn, Co, Ge, Ag, Cd and Cu loaded in PC1 are enriched and correlated in deep section along depth of orebody whereas In, Ga loaded in PC2 are enriched and correlated in the whole orebody. See text for explanation.

almost low (~1 ppm, with several samples up to 25 ppm), which indicates a narrow range of Sn concentrations across the dataset. From the Daliangzi deposit to MVT, skarn type, and massive sulfide deposits, the concentration of Sn ranges from low to high levels, which also confirms the important role of magma processes in enriching tin (Fig. 4(K)). The Jinding type deposit has the lowest Sn concentration in comparison with the deposits collected in Electronic Appendix A. The spectra of Sn are usually ragged, whether or not the sphalerite contains inclusions (Fig. 5(D) and (F)), with local outliers up to 25 ppm, which proves that Sn in sphalerite is incorporated via microscopic inclusions. Tin concentration exhibits a weak positive correlation with In (Fig. 6(O)).

In addition, gold concentrations from the Daliangzi deposit are generally below the minimum detection limit. Uranium and thorium concentrations are ≪1 ppm in all analyzed sphalerite.

6.2. *Principal component analysis of minor and trace elements*

The results of the PCA applied to the LA-ICP-MS dataset are shown in Fig. 7, where elements and spot analyses are projected on the PC1 vs. PC2 plane, accounting for 63% of element content variability. Element distributions (Fig. 7(B)) highlight two main groups or element correlation clusters. One group is composed of Fe, Mn, Cu, and trace elements Co, Ge, Cd, and Ag (loading PC1), while the other group is composed of In and Ga (loading PC2). In PC1, siderophile elements (Fe and Mn) have opposite signs to the chalcophiles (Cu, Cd, and Ag). Based on such a situation, PC1 is divided into three subgroups: PC1-1 (Fe and Mn), PC1-2 (Co and Ge), and PC1-3 (Cu, Cd, and Ag). PC1-3 occupies an intermediate positon between the Fe, Mn, and Cu plus trace elements group along the PC1 and PC2 (In and Ga) groups. Spot representations in the PC1 vs. PC2 plane (Fig. 7(A)) refer to the sample locations described in Section 4 ranging from the 1884 m to 2064 level of No. I orebody. In detail, PC1 (loaded by Fe, Mn, Co, Cu, Ge, Ag, and Cd), which includes three subgroups (PC1-1, PC1-2, and PC1-3), is correlated with the samples collected in the lower orebody (< 1944 m level), whereas PC1 and PC2 (loaded by In and Ga) are correlated with those in the upper orebody (> 1944 m level). Specially, PC2 (loaded by In and Ga) is correlated with the highest volume of sample locations in orebody No. I. According to Fig. 7, it is obvious that different sampling locations, representing different geological processes of element incorporation during mineral growth, do have a great influence on the composition of associated minerals, for example, sphalerite.

7. Discussion

7.1. *Element clustering in zoning patterns*

The appearance of multielement correlation groups specific to sampling locations identified by using PCA (Fig. 7), i.e., Fe, Mn, Cu, Ag, Cd, Co, and Ge (loading PC1) in the lower orebody and Fe, Mn, Cu, Ag, Cd, Co, Ge, Ga, and In (loading both PC1 and PC2) in the upper orebody, may be reasonably linked to different processes of incorporation during sphalerite growth. It is common for the local physicochemical conditions to be quite different from the core to the rim of a huge orebody, which can result in large variations in mineral compositions. Zoning patterns indicated by different compositions of ores are generally considered to be related to self-organized nonequilibrium processes occurring during growth of a huge orebody under specific conditions, as said by Di Benedetto et al. (2005) and Belissont et al. (2014) to zoned sphalerite crystal. Typically, large-scale variations of the mineralizing fluid composition in open systems and microscale variations of the fluid composition in closed systems are involved in the formation of a huge orebody. Microscale variations are influenced and controlled, to some extent, by the large-scale variations during a mineralization event. Numerous indexes of such variations, such as Zn/Cd, Fe/Cd, In/Zn, In/Fe, Co/Ni, and In/Ge ratios, are used to limit the environmental parameters (Cook et al., 2009; Ye et al., 2011; Murakami and Ishihara, 2013). Element clustering caused and constrained by the surrounding setting provides a “window” for studying the properties of the surroundings.

The obvious trends of enrichment of minor and trace elements in sphalerite from different depths of orebody No. I show clear laws of spatial distribution. Samples enriched in Fe, Mn, Co, Cu, Ge, Ag, and Cd occur in the lower orebody whereas those high in Fe, Mn, Co, Cu, Ge, Ag, Cd, Ga, and In appear in the upper part. Except Ga and In, the other elements is the same. This differentiation was probably triggered by the decrease in temperature with decreasing depth as measured from the earth's surface. This speculation is based on the obvious Ga enrichment relative to In (high Ga/In ratios) in one mineral species (e.g., sphalerite) being attributed to the low-*T* setting as for a given orebody (e.g., Frenzel et al., 2016, and references therein). Such a situation relates to the crystallization habit that Ga concentrates in a lower-temperature hydrothermal setting, whereas In clusteres in a hypo-mesothermal hydrothermal environment (Cook et al., 2009). This case is consistent with the micothermometry result that most sphalerite were formed in a range between 150 °C and 200 °C being in a lower-temperature zone (Zheng and Wang, 1991). Therefore, temperature does play a great role in controlling the zoning pattern of huge orebodies, for example, orebody No. I, with Ga enrichment in the top at the Daliangzi deposit. However, based on our limited evidence, pH, redox conditions, salinity, and the availability of ligands to form choride, sulfide, or hydroxide complexes of the ore-forming fluids cannot be ruled out as potential controls.

7.2. Correlation trends and substitution mechanisms

Numerous authors have completed laboratory work on the substitution mechanisms in sphalerite (e.g., Di Benedetto et al., 2005; Cook et al., 2009, 2015; Belissont et al., 2014; Frenzel et al., 2016; George et al., 2016); however, much work remains to be done on the geochemical complexity and genetic implication represented by the special concentration of elements inside. As we know, simple substitutions have been discovered between Zn and other divalent cations, for example, Zn^{2+} being replaced by Cd^{2+}, Mn^{2+}, Co^{2+}, Ni^{2+}, Hg^{2+}, and Sn^{2+} (Seifert and Sandmann, 2006; Murakami and Ishihara, 2013), but some more complicated substitution mechanisms also occur in the structure of sphalerite independent of the simple substitutions above (Belissont et al., 2014; Frenzel et al., 2016; George et al., 2016, and references therein). Comparison using statistical diagrams of trace element concentrations in sphalerite reveals distinctive positive and negative correlations between them, proving the existence of more complicated substitution mechanisms. Substitution mechanisms with the elements trapped can be used both for understanding the mineralization process and for identification of the genetic type of a given deposit (Ye et al., 2011; Belissont et al., 2014; Frenzel et al., 2016, and references therein). The Daliangzi deposit has a very close relationship with carbon (including bitumen, plumbago, etc.), which occurs extensively around or inside the orebodies. These carbon might have provided the reducing agent indispensable for reduction reaction which caused the deposition of most metals. The presence of large amounts of carbon built a special ore-forming environment with a corresponding special metallogenic process therein.

Sphalerite from the Daliangzi deposit is more enriched in iron than any other element. The substitution mechanism of Zn in sphalerite caused by Fe is of great importance to sphalerite and to the Daliangzi deposit itself. Across the sample suite, it is difficult to identify pyrite incorporated by sphalerite with a microscope, and, especially, combined with the smooth profiles usually (Fig. 5(C) and (E)), these would indicate that Fe is generally present as a kind of solid solution. A simple substitution mechanism between Zn^{2+} and Fe^{2+} is thus twice verified and is the most important one for sphalerite itself since iron is the most ubiquitous minor element in sphalerite. Plotting individual elements against Fe should, therefore, indicate whether presence of iron assists incorporation of other components as suggested by Cook et al. (2009). A broad positive correlation is observed between Fe and Mn, Cd, and Ge across the dataset (Fig. 6(C) and (E)), with an inconspicuously positive correlation between Fe and Co, supporting the idea mentioned above. Moreover, the mechanism of Zn^{2+} replaced by Mn^{2+}, Cd^{2+}, Co^{2+}, and Ge^{2-4+} did play an important role in sphalerite, whereas Zn^{2+} replaced by other divalent cations seems to be not so important for the low contents across the dataset despite their significant ore-genetic implications. In contrast to such a situation, some suggestions of an anitcorrelation between Fe and Mn are still widely acceptable (Di Benedetto et al., 2005), indicating the elusive complexity therein.

Despite germanium and gallium being very close to each other in the periodic table of elements, only a weak positive correlation can be detected in the Daliangzi deposit (Fig. 6(I)). Identifying the substitution mechanisms of these elements in sphalerite has been considered to be elusive (Belissont et al., 2016). In the total dataset, Ge shows a clear positive correlation with Fe content (Fig. 6(E)), with an unobvious positive correlation with Ga, suggesting that Fe-enriched sphalerite carries a higher level of both elements. Germanium always exhibits both Ge^{2+} and Ge^{4+} oxidation states, with Ge^{4+} being the more common oxidation state, and thus it may be expected to mimic iron in terms of substitution in sulfides (Ye et al., 2011). Gallium usually forms Ga^{2+} and Ga^{3+} ions, with Ga^{3+} being ubiquitous; the substitution mechanism relies on the formation environment nearby. The weak positive correlation between Fe and Ge, with a slope of nearly 2 (Fig. 6(E)), raises the possibility of a coupled substitution of $4Zn^{2+} \leftrightarrow 2Fe^{2+} + Ge^{4+} + \square$ (where $\square$ denotes a vacancy). However, a simple substitution can also occur when Ge is in a reduced state (Ge^{2+}), so more supporting evidence is needed to corroborate such a speculation. Many sphalerite samples from the Saint-Salvy vein-type Zn–Ge–Ag deposit support the reality of a proposed coupled substitution of $3Zn^{2+} \leftrightarrow 2Cu^{+} + Ge^{4+}$ according to the good correlation with a slope ≈ 2 between Cu and Ge contents (Belissont et al., 2016). Such a situation may also occur in sphalerite samples from the Daliangzi deposit (Fig. 6(A)). However, another weak positive correlation between Ge and Ag contents (Fig. 6(J)) probably supports the coupled substitution mechanism of $3Zn^{2+} \leftrightarrow 2Ag^{+} + Ge^{4+}$. As for Ga, despite its having approximately the same geochemical properties as Ge, those trends showed above by Ge seem to be inconspicuous.

By comparing the contents listed in Electronic Appendix A, it can be easily seen that silver and antimony are markedly enriched in sphalerite samples with a correlation of 1:1 (Fig. 6(L)). This suggests a possible substitution mechanism of $2Zn^{2+} \leftrightarrow Ag^{+} + Sb^{3+}$ and a combination of lattice-bound Ag and Sb, which assist the corporation of solid solution alongside some microscopic inclusions of Ag- and Sb-bearing minerals. Also, there is a clear positive correlation between Sb and Cu with a slope of nearly 1 (Fig. 6(B)), and a possible substitution mechanism of $2Zn^{2+} \leftrightarrow Cu^{+} + Sb^{3+}$ can also be indicated. This makes the lattice-bound Cu and Sb believable.

A weak positive correlation is observed between As and Tl contents across the dataset, with a slope of nearly 3 (Fig. 6(K)). This trend has been investigated exclusively among the deposits collected from other types, which also feature the mineralization occurring in the Daliangzi deposit. Given that As- and Tl-bearing mineral inclusions were recognized in many of the time-resolved depth profiles, it is reasonable to believe that fangite (Tl_3AsS_4) became incorporated in sphalerite as independent minerals (Ye et al., 2011).

Nickel and cobalt, being close to each other in the periodic table of elements, show similar geochemistry and often correlate with one another. Co and Ni show a broad positive correlation in the total dataset, with a reasonable slope of Co/Ni ≈ 1 (Fig. 6(H)). This trend appears to be independent of any potential Fe–Co or Fe–Ni relationships. A simple substitution mechanism of $Zn^{2+} \leftrightarrow Co^{2+}$ or Ni^{2+} is thought to be widely occurring (Cook et al., 2009), which probably fits the sphalerite cases from the Daliangzi deposit. In other types of deposits, e.g., skarn type and special Jinding deposits, cobalt is much more enriched than nickel (Fig. 6(H)), which implies an important ore-genetic significance.

The weak positive correlation between In and Sn, with a probable slope of 1 (Fig. 6(L)), suggests a potential coupled substitution of $3Zn^{2+} \leftrightarrow In^{3+} + Sn^{3+} + \square$. Such a speculation is based on In occurring in a trivalent state in most materials (Epple et al., 2000) and Sn being present in a trivalent state to match this probable coupled substitution, as suggested by Belissont et al. (2014). However, we cannot preclude Sn^{2+} and Sn^{4+} (being the main states) from also existing in the lattice of sphalerite because of the paucity of solid evidence. This raises the possible coupled substitution of $3Zn^{2+} \leftrightarrow In^{3+} + Sn^{2+} + (Cu, Ag)^{+}$ or $4Zn^{2+} \leftrightarrow In^{3+} + Sn^{4+} + (Cu, Ag)^{+} + \square$. The matter, not only for Sn, cannot be resolved until suitable methods are adopted to fix the oxidation state, for example, using X-ray absorption or μ-X-ray absorption near edge structure/extended X-ray absorption fine structure methods as suggested in Pattrick et al. (1998), Goh et al. (2006), and more recently in Cook et al. (2012).

Fig. 8 shows some selected (As + Sb) vs. (Cu + Ag) and also Pb data to investigate the relationships that are indispensable for the

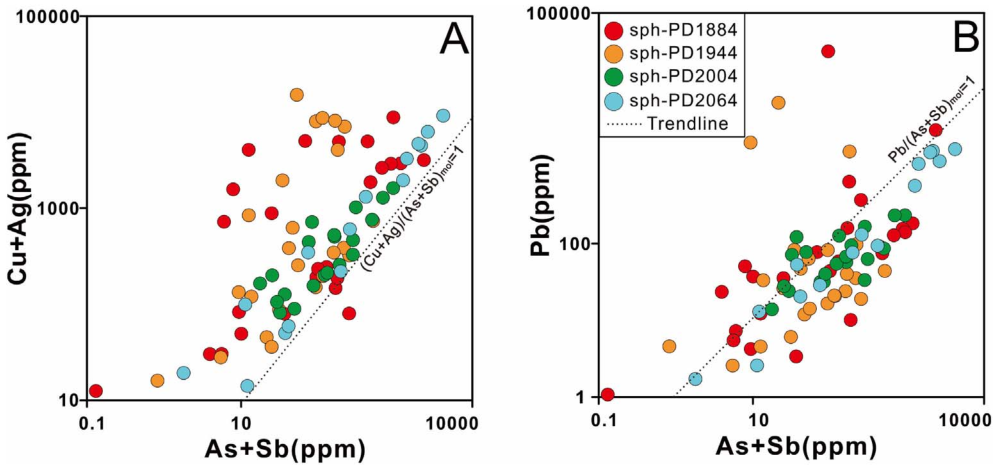

Fig. 8. Correlation plots of (A) (Cu + Ag) vs. (As + Sb) and (B) Pb vs. (As + Sb) in sphalerite from the Daliangzi deposit. Trendlines show correlation of 1:1. Note in A) that typical elements (Cu + Ag) and (As + Sb) are for tennantite-tetrahedrite solid solution; typical elements Pb and (As + Sb) are for jordanite-geocronite solid solution.

tetrahedrite–tennantite and jordanite–geocronite solid solution series in sphalerite. The obvious positive correlations shown in Fig. 8(A) and (B) suggest the probable occurrence of the solid solution series mentioned above in sphalerite from the Daliangzi deposit. Moreover, it is worth emphasizing that these kinds of solid solutions, which differ from those in sphalerite from other genetic types, are important distinct signatures formed by the special mineralization setting in the Daliangzi deposit. In contrast, the solid solution series from sphalerite samples in the vein-type Toyoha deposit, Japan, are typically the Zn–(Cu + Ag)–(Sn–In) system composed principally of sphalerite–$CuZn_2InS_4$, roquesite–-$CuZn_2InS_4$, chalcopyrite–stannite, etc. (Ohta, 1995; Cook et al., 2009).

In addition, according to Fig. 6, some distinct geochemical signatures in terms of source and genetic type can be obtained. For example, sphalerite from massive sulfide deposits shows enrichment in In, Sn, Fe, and Mn. Those from skarn type deposits are high in Fe, Mn, and Co. The Jinding deposit has much more Tl in sphalerite. MVT deposits feature high Ga, Ge, Ag, and Sb contents in sphalerite. These characteristics highlight the differences caused by various mineralization processes, suggesting that the geochemistry of typical minerals has great potential for discriminating ore-forming processes and identifying ore types.

7.3. Significance of PCA in sphalerite-bearing ore deposit types

Dozens of authors have suggested that the type of ore deposit has a great influence on the minor and trace element composition of sphalerite (e.g., Oftedahl, 1940; Cook et al., 2009, 2015; Ye et al., 2011; Belissont et al., 2014; Frenzel et al., 2016). Especially, those factors indispensable for a given deposit—such as the source of ore components, features of the ore-forming fluid and evolutionary history, wall rocks, sites and mechanisms of deposition, as well as the relative volume of sphalerite within the orebody—have a close relationship with the mineralization type, which will more or less control the elemental fractionation and the overall concentrations in sphalerite. We combed our LA-ICP-MS minor and trace element dataset from the Daliangzi deposit with the dataset of Ye et al. (2011) and conducted a multivariate statistical PCA to trace the factors above. It is worth noting that Ye et al. (2011) collected data from different kinds of ore type but mainly from the same tectonic unit (the Yangtze continent) as well as the Daliangzi deposit. In this study, we also attempt to construct a geochemical framework to discriminate between genetic types of ore deposits located in this tectonic unit.

To clearly show the different trends or variabilities hiding behind the multiple dimension dataset of minor and trace elements, the mean values of specimens collected from Ye et al. (2011) are adopted for PCA. The LA-ICP-MS data of sphalerite from the Daliangzi deposit with those mean values are compiled into a large database (Electronic Appendix A), log-transformed, and then processed by PCA only on the minor and trace element contents selected (Fe, Mn, Cu, Cd, Ga, Ge, In, Co, and Ag). The aim is to determine whether minor and trace element features in sphalerite can efficiently be used to discriminate between ore genetic types. Note that these variabilities are dominated by the Daliangzi deposit at a high level because of the mean values used, suggesting that there are numerous possible treatments other than the one introduced here. The Daliangzi deposit dataset, having strong microscale compositional variations, has been plotted together with those mentioned above in the PCA to focus the analysis on variations of minor and trace elements in sphalerite at the scale of the ore deposit (i.e., the scale of Ye et al.'s, 2011 study).

The PCA results, according to Fig. 9, show that factors PC1* and PC2*, accounting for 63% of element content variability, are the most important enrichment trends throughout the whole database. The first factor of discrimination (PC1*) is mainly loaded with Cu, Ag, Ga, and Ge whereas the second order of discrimination (PC2*) is primarily loaded with Fe, Mn, and Co. Cu and Cd occupy an intermediate position between these factors (Fig. 9(A)). Namely, high Fe, Mn, and Co contents in sphalerite may typify magmatically driven hydrothermal systems (i.e., skarn and VMS deposits; Fig. 9(B) and (C)). They are chiefly found in the top section of the graph. In contrast, the group of Cu, Ag, Ga, and Ge tends to be enriched in low-*T* deposits having nothing to do with a magmatic source (i.e., MVT deposits and the special Jinding deposit; Fig. 9(B) and (C)) according to Ye et al.'s (2011) petrogenetic classification, although special signatures do exist reflecting local geological settings. These are located in the bottom section of the plot. Thus, this framework is likely to illustrate a positive temperature gradient and differences in metal sources from the bottom to the top of the plot (Fig. 9(C)). Therefore, it is clear that the differences among deposits follow a simple one-dimensional pattern. That is, the trace element contents of sphalerite from different deposit types only differ substantially along one of the nine dimensions of the sample space considered here, and the dimension, exclusively, is described by PC2* (Fig. 9(C)).

As suggested by Cook et al. (2009), Ye et al. (2011), Belissont et al. (2014), and Frenzel et al. (2016), a given deposit or even an ore province limits enrichment signatures of minor and trace elements. The sphalerite from the Daliangzi deposit occupy the bottom of the plot (Fig. 9(C)), consistent with the low-*T* deposition and the nonmagmatic nature of the depositional fluids (Zheng and Wang, 1991). Finally, to be proved once again, even if large variations in the trace element contents occur in the sphalerite of a given deposit, trends of minor and trace element distributions really show petrogenetic significance and may provide discriminators for identifying different genetic types of given deposits.

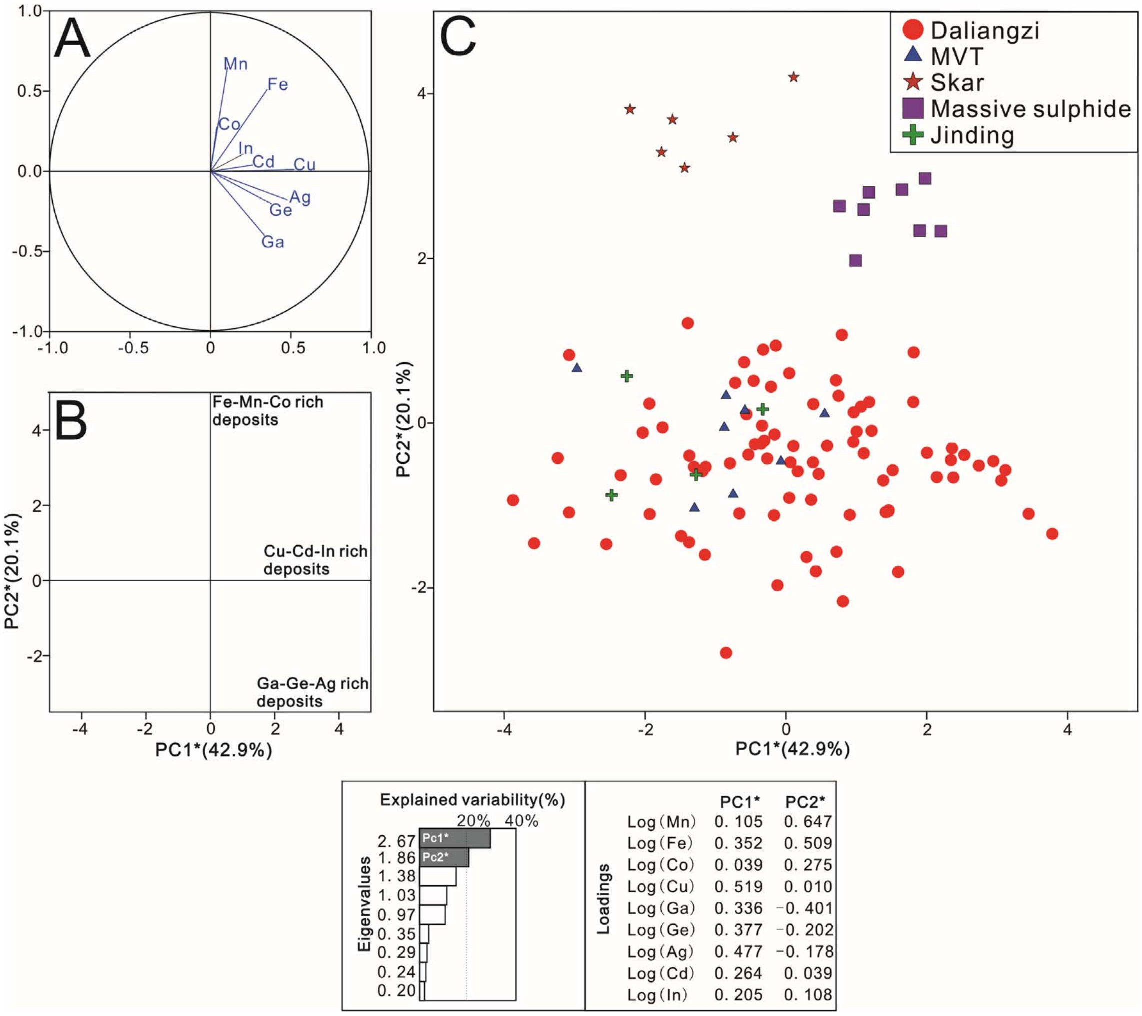

Fig. 9. PCA of Daliangzi deposit and Ye et al. (2011) LA-ICP-MS dataset of sphalerite from different ore deposits of various types in South China. Eigenvalues and loadings of the PCA are indicated. (A) Plot of the elements (variables), (B) interpretative scheme in terms of element enrichment for spot analysis (individuals), and (C) the scatterplot of the log-transformed spot analysis datasets (individuals) of the above publications including Daliangzi datasets, plotted in the PC1* vs. PC2* plane. This PCA of the trace element contents in sphalerite from various deposits confirm their geochemical significance in discriminating between genetic ores types (e.g., high Fe, Mn, Co contents for magmatic-related deposits, and Cu, Ag, Ga, Ge for low-temperature deposits). In the bottom insert, eigenvalues, eigenvector loadings are shown.

7.4. Metallogenic consideration

No. I orebody in Daliangzi Pb-Zn deposit, being of super-large scale, the mineralzation process arises the extensive attention of readers. Particularly, the metallogenic periods and stages of which need to be well investigated. This whole body, from the top to the bottom, the minor and trace element distribution patterns only show tiny difference indicated by the enrichment of Ga. Such a situation can reasonably be ascribed to the decreased temperature which dominated the zoning pattern. For a given orebody, temperature should play a more important role in controlling the distribution pattern of minor and trace elements rather than other factors, such as the metal source, deposit type, and even the proportion of sphalerite in the ore, as proposed by Bethke and Borton (1971) and Belissont et al. (2014). Therefore, we didn't observe any odds which might have been produced by another ore-forming process. The similar distribution patterns of trace elements in sphalerite (except Ga) at four different horizons indicated by Figs. 7 and 8 mineral-geochemically support the probability that the formation of orebody No. I resulted from a single ore-forming event.

Compared to other deposits of various types including VMS, SEDEX, Skarn, Jinding, and even MVT, the Daliangzi Pb–Zn deposit is known not only for its huge scale of mineralization but also but also for its greater enrichment of Cu, Cd, Ga, Ge, Ag, Sb, and Tl (Figs. 4 and 6), which is considered to be related to a low-temperature metallogenic process (Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014; George et al., 2016).

Accurately confirming the mineralization process and metallogenic type of a given deposit is crucial in theoretical research and industrial exploitation in complex geological situations. For a given element, its concentration and fractionation in sphalerite have significant implications for the type of deposit, ore-forming temperature, metal sources, cooling process, erosion levels, etc. (Cook et al., 2009; Zhang et al., 2014). Statistical methods, such as PCA performed on trace element contents, can establish a valuable framework (Figs. 7 and 9) not only to reveal the hydrothermal evolution but also to provide geochemical evidence for differentiating metallogenic types.

In conclusion, studies focusing on the variations of minor and trace element distributions are extremely useful for understanding the mineralization process and provide a theoretical foundation for geochemical tools or index systems for identifying and exploiting deposits.

8. Conclusions

This integrated study coupling mineralogical observation and in situ LA-ICP-MS analysis of minor and trace elements has allowed an investigation of their distribution, substitution mechanisms, and interrelationships and has provided implications for geochemical signatures of sphalerite-bearing ore deposits. The main conclusions are as follows:

1. It is reasonable to conclude that orebody No. I, being super-large, was formed by only one ore-forming event as can be discerned from the similar enrichment laws of most trace elements trapped in sphalerite of different depths.
2. Sphalerite samples enriched in Fe, Mn, Co, Cu, Ge, Ag, and Cd occur in the lower orebody while those high in Fe, Mn, Co, Cu, Ge, Ag, Cd, Ga, and In are present in the upper orebody, suggesting that the temperature gradient controls this inconspicuous difference in trace element enrichment, which was ascribed to the preference of Ga enrichment (high Ga/In ratios) in a relatively low temperature setting.
3. The enrichment of Fe assists incorporation of many trace elements in sphalerite form orebody No. I of the Daliangzi deposit, such as Mn, Cd, Co, Ge, and (Ag + Sb). Some notable coupled substitutions are also indicated: $4Zn^{2+} \leftrightarrow 2Fe^{2+} + Ge^{4+} + \square$ (where $\square$ denotes a vacancy), $2Zn^{2+} \leftrightarrow Ag^{+}$ $(Cu^{+}) + Sb^{3+}$, and $2Zn^{2+} \leftrightarrow 2Ag^{+}$ $(Cu^{+}) + Ge^{4+}$. Moreover, tetrahedrite–tennantite and jordanite-geocronite solid solution series, indicated by the (As + Sb) vs. (Cu + Ag) or (As + Sb) vs. Pb binary scatter plots, are considered to be the important distinct signatures formed by the special metallogenic setting of Daliangzi Pb-Zn deposit.
4. The outstanding enrichments of Cu, Ga, As, Cg, Ge, Ag, Sb, Pb, Ni in sphalerite from different depths of the super-large No. I orebogy mutually imply a lower-temperature hydrothermal process dominating the mineralization in Daliangzi Pb-Zn deposit.
5. A framework established by PCA has been proposed to study the base-metal deposit types and the relevant metallogenic processes typical for those deposits located on the Yangtze continent in South China.

Finally, a yet-to-be-identified suitable quantitative index of these characteristic elements (Fe, Mn, Co, Ag, Ga, and Ge) can be used for not only identifying mineralization processes but also for providing tools to investigate the erosion level of a given orebody or deposit. This poses interesting questions for future research work. Otherwise, the nature of the controlling factors behind the other principal components in our dataset still awaits identification.

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Conflict of interest

The authors declear no conflict interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gexplo.2017.12.014.

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